

Dirac Delta-Based Applications vs. Dual-Purpose Agent Deployment Design

Introduction

Signal-multiplexing systems with multiple intelligent agents must balance *deterministic* tasks and *uncertain* public-utility workloads. The user's **Agent Deployment Design** outlines a dynamic allocation framework where agents (and their large-language-model (LLM) strengths) are optimally routed across channels using online learning of task complexity and feedback-driven adaptation. To better situate this design within existing research, this report reviews how the **Dirac delta function** has been employed in multi-agent coordination and links those applications to the dual-purpose plan. The Dirac delta, denoted $\delta(x)$, is a generalized function that is zero everywhere except at zero and integrates to one. In multi-agent systems it often models impulsive events, sampling operations or distributions of agent locations.

Overview of the Dual-Purpose Agent Deployment Plan

The uploaded document (`Agent_Deployment_Design.md`) proposes a sophisticated framework for routing agents across two channels: **specific function signals** (deterministic tasks) and **public utility signals** (tasks with uncertain complexity). Key features include:

Element	Description	Purpose
Decision variables	Number of agents N_i per channel, LLM strength S_i , and routing fractions R_{ij}	Manipulate allocation across channels and adjust computational power
Learnable costs	Complexity functions for specific and public signals ($C_{\text{specific}}(i)$, $C_{\text{public}}(i)$) learned via online gradient descent	Adapt resource allocation to observed workloads
Objective	Minimizes a cost combining workload, LLM strength penalties and performance rewards, subject to capacity and bandwidth constraints	Achieves efficient processing while incentivizing high throughput and low latency
Adaptive loop	Periodically observes queue depths and performance, updates complexity estimates, formulates and solves an optimization problem, re-allocates agents and LLM strengths, and updates learning parameters	Enables real-time adaptation to changing signal patterns and bandwidth
Dual-path processing	Separate assignment formulas for specific and public signals; public assignments account for uncertainty via variance terms	Balances deterministic processing with reasoning-intensive tasks

In essence, the design treats agents as *computational resources* whose number and strength are manipulated to maximize throughput subject to complexity and bandwidth; complexity models are learned online and performance metrics provide feedback. The plan currently operates in discrete optimization cycles (e.g., every 100 ms) without explicit mention of impulsive actions or sampling-based measures.

Dirac Delta Applications in Multi-Agent Coordination

Research literature uses the Dirac delta function to model instantaneous events, represent agents as point masses, or define time-averaged statistics. Below are key applications related to multi-agent coordination.

Impulsive Consensus Control

Quantized consensus with nonlinear dynamics. To achieve consensus when agents communicate quantized states, Zhu, Zheng & Wang (2015) designed an *impulsive control input* that applies corrections only at discrete times t_k . The control for agent i is

$$u_i(t) = - \sum_{k=1}^{\infty} b_k \sum_{j=1}^N a_{ij} Q(x_i(t) - x_j(t)) \delta(t - t_k), \quad (1)$$

where $Q(\cdot)$ is a quantization function, a_{ij} denotes adjacency weights and $\delta(t - t_k)$ is the Dirac delta. Because $\delta(t - t_k) = 0$ except at the impulse times, the control is zero between impulses and produces instantaneous jumps at t_k . Using this impulsive formulation, the authors derive conditions on the gains b_k and impulse intervals that guarantee consensus ¹. The Dirac function therefore models discrete control actions that update the system state at isolated instants.

Ergodic Coverage and Search

Ergodic metrics for coverage control. Ergodic coverage aims to drive agents' trajectories so that they spend time in proportion to a target spatial distribution. The *time-averaged statistics* of trajectories are defined using the Dirac delta. For N agents with trajectories γ_j , the spatial time-average distribution is

$$C_t(x) = \frac{1}{Nt} \sum_{j=1}^N \int_0^t \delta(x - \gamma_j(\tau)) d\tau, \quad (2)$$

where $\delta(x - \gamma_j(\tau))$ measures how much time the agent spends at position x . The ergodic metric compares the Fourier coefficients of this distribution to those of a target distribution. In energy-aware ergodic search, a similar definition appears: the *time-average distribution* $h_{q(t)}$ over a window $[t_0, t_f]$ is defined via $\int_{t_0}^{t_f} \Delta_{q(t)} dt / (t_f - t_0)$, where Δ is a Dirac delta ². These formulations use Dirac deltas to sample the positions along trajectories and are central to computing ergodic control laws.

Multi-agent ergodic coverage with obstacles. Salman et al. (2017) extend ergodic coverage to domains with obstacles. Their ergodic metric (Eq. (1)) uses the Dirac delta as in (2) to compute $C_t(x)$, and control laws steer agents based on gradients of this metric. The Dirac delta thus serves as a *sampling operator* converting continuous trajectories into spatial distributions.

Geometric Task Allocation

Schwager’s **distributed and decentralized geometric task allocation** frames task allocation for a swarm by defining an error functional

$$G(q) = \frac{1}{2} \iint (\Phi(x, y) - \sum_{i=1}^N \delta(x - x_i) \delta(y - y_i))^2 dx dy, \quad (3)$$

where $\Phi(x, y)$ is a spatial demand profile and $\delta(x - x_i) \delta(y - y_i)$ represent agents as Dirac point masses. The Dirac distribution enables the integral to capture discrete agent positions within a continuous domain. Taking functional derivatives yields gradient-descent motions that reposition agents to satisfy the demand profile. This technique exemplifies how Dirac deltas represent *agent presence* in coverage or allocation problems.

Continuous-Time Planning with Temporal Constraints

In continuous-time Decentralized Markov Decision Processes (DEC-MDPs), agents may need to *wait* to synchronize actions. Nair et al. approximate probability density functions of wait times using piecewise constant functions plus Dirac delta functions to capture instantaneous spikes at zero wait. The policy update equation includes terms like $\delta(t - 0.25)$, modeling instantaneous events within continuous probability densities ³.

Coupling Dirac-Based Concepts with the Dual-Purpose Plan

Impulsive Controls vs. Periodic Optimization

The quantized consensus and asynchronous impulsive control literature uses Dirac delta inputs to apply **instantaneous corrections** at discrete times. In the user’s plan, agent allocation and routing decisions are updated periodically (every 100 ms) by solving an optimization problem. While the plan does not currently model impulses, it could benefit from *impulsive adjustments* in extreme scenarios. For example, when queue depths of public signals spike or bandwidth drops abruptly, an impulsive reallocation—analogue to applying a Dirac delta update—could instantaneously move agents to critical channels without waiting for the next optimization cycle. Designing such impulsive corrections would require deriving conditions under which stability and performance are maintained, similar to those in the consensus literature ¹.

Sampling-Based Coverage Metrics for Signal Queues

The ergodic coverage framework introduces **time-averaged statistics** via Dirac deltas to evaluate how agents explore a spatial domain. In the dual-purpose design, analogous metrics could be defined over **queue states** instead of physical space. For instance, let $q_i(t)$ be the queue depth of channel i ; one could define a time-average distribution over queues:

$$C_t(q) = \frac{1}{Nt} \sum_{j=1}^N \int_0^t \delta(q - q_i(\tau)) d\tau.$$

Comparing this empirical distribution to a target distribution (e.g., balanced workloads) yields an **ergodic metric** that can be included in the optimization objective. Agents would be routed so that queue depths sample the desired distribution over time, ensuring balanced processing. This perspective brings the ergodic approach into the domain of signal-multiplexing.

Representing Agents as Dirac Point Masses

In geometric task allocation, Dirac deltas represent agents as point masses in a continuous demand field. Similarly, the dual-purpose plan could interpret **each agent's service capacity** as a Dirac point in an abstract capacity space. The optimization could then minimize a functional akin to (3), comparing the current distribution of agent capacities across channels to the desired demand distribution. This continuous representation might facilitate gradient-based allocation rather than discrete optimization, enabling smoother adaptation when agent counts are large.

Modeling Instantaneous Events in Learning Dynamics

Continuous-time planning work uses Dirac deltas to represent instantaneous transitions or wait-time spikes ³. In the dual-purpose plan, *learning events*—such as updates to complexity models upon receiving new data—are currently treated as continuous gradient steps. If certain signal types induce abrupt changes in complexity (e.g., a sudden emergence of a new public-utility task), these could be modeled as **Dirac impulses** in the complexity update equations. The learning rate could include impulse terms to quickly incorporate high-impact observations.

Discussion and Future Directions

Coupling Dirac-based techniques with the user's dual-purpose deployment framework reveals several research opportunities:

1. **Impulsive reallocation policies.** Develop impulsive control laws for agent redistribution that trigger when queue depths or complexity errors exceed thresholds. Stability conditions similar to those in consensus problems ¹ could ensure that impulsive reallocations do not destabilize the system.
2. **Ergodic metrics over abstract spaces.** Apply ergodic coverage concepts to non-spatial variables such as queue depths or complexity estimates. Using Dirac deltas as sampling operators, define time-averaged distributions and incorporate them into the cost function to achieve balanced processing.
3. **Continuous representation of agent allocation.** Model the allocation of agents as a continuous density over channels using Dirac deltas and minimize an error functional analogous to geometric task allocation. This could enable gradient-based rather than combinatorial optimization.
4. **Impulsive learning updates.** When observing atypical signals that dramatically update complexity models, incorporate Dirac impulses into the learning dynamics. These impulses could rapidly adjust complexity estimates or learning rates when sudden changes occur.
5. **Integration with energy and resource constraints.** The energy-aware ergodic search literature demonstrates how constraints (e.g., battery level) can be integrated into coverage control ². Similar ideas could be used to handle bandwidth and processing-power constraints in the dual-purpose design, using Dirac delta formulations to represent time-averaged resource utilization.

Conclusion

The Dirac delta function, though an abstract mathematical object, plays a tangible role in modeling **impulsive events**, **sampling operations**, and **point-mass representations** in multi-agent coordination. By examining impulsive consensus control, ergodic coverage, geometric task allocation, and continuous-time planning, this report highlights how Dirac-based formulations can inform the design of dynamic agent routing and resource allocation systems. Integrating impulsive control laws and sampling-based metrics into the dual-purpose agent deployment plan could enhance responsiveness to sudden changes, enable more flexible optimization, and establish a richer theoretical foundation for signal multiplexing in multi-agent systems.

1 Quantised consensus of multi-agent systems with nonlinear dynamics

https://web.xidian.edu.cn/yszhenf/files/20181227_124211.pdf

2 Seewald_ICRA2024_2.pdf

https://www.eng.yale.edu/grablab/pubs/Seewald_ICRA2024_2.pdf

3 Continuous Time Planning for Multiagent Teams with Temporal Constraints

<https://www.ijcai.org/Proceedings/11/Papers/085.pdf>